

Source-to-effect trace: bounded export under A4 ENFORCE

LA-015 identity 104729:A4:family:10df17dbfee02e86d083e2784e7aaec79a4691d1441d1ceff4ef944f2695b949:case-0 revision ee6d73c4a5fc30d0
 case family:10df17dbfee02e86d083e2784e7aaec79a4691d1441d1ceff4ef944f2695b949:case-0 population=skill mutation=none decision=allow useful_work=True effects=1
 Each row is a retained transition for this attempt. Absent/partial steps are labeled; SAT is satisfiability, not a kernel theorem.

Appendix D twelve-step trace (permitted useful work)

Stp	Status	Method	Observed evidence
1	ran	Resolve source and declaration manifests by exact revision/CID	Frozen source_id skill:cde85f1fbfa19461702271bc52c9b65be4eb944e75e427a2470e9f34227dd693 with content identity ef917528b31bf690eda2fac57a9f568f7f66f15a99384d00447e12fdeb40e7e0; IPFS publication was not performed.
2	ran	Import skill procedure as data; derive or validate IntentIR	SkillCenter normalizer produced skill_intent_ir document_id=intent:skillcenter:8e128cbf4c4daa54d7d61c54864dd1759b03d3279f5379611420f54dd9397967 schema_valid=True. Typed compiler available=False.
3	partial	Select applicable LegallR and reviewed SecurityIR constraints	LegallRCompilerAPI was not the selected adapter. Independent human legal/security review was not used. Machine-contract span linkage ran.
4	ran	Bind actual requested tool, arguments, actor and expected effects	Bound actor=did:client:tenant-a tool=export_json path=exports/fa-075d8327ad20c2d4.json candidate_digest=b0875e6b1a86ac83.
5	partial	Correlate intent-side effects with code/handler-side observations	The handler accepts a declarative export_json instruction, not generated source. LA-016 produced 0 generated programs. Declared filesystem.export_json effects were compared with the independent journal observer.
6	ran	Compose and execute the required formal jobs; retain non-successes	QF_BOOL SAT provider=unsat checker agrees=True. Authority is satisfiability, not theorem_proof.
7	ran	Produce an exact-context decision, not an effect	An internal BoundContext decision receipt and capability were derived for the exact actor/tool/args digest. Harness selected_evidence_cids are constants, not a published proof.
8	ran	Check signed delegation and host-controlled current policy/state	Real Ed25519 UCAN verifier decision=allow token_sha256=7c3c672fef671199 signature_present=True.
9	ran	In enforce mode, revalidate and consume the exact permitted use	ENFORCE decision=allow store=duckdb-file-typed-quack-owner in_memory=False consume=accepted.
10	ran	Dispatch once through the selected protected path	BoundedExportHandler.export_json handler_calls=1 delegate_called=True.
11	ran	Record observed outcome separately from proof and authorization	Independent observer v2 counted effects=1 journal=1 useful_work=True (LA-015 compact useful_work=True).
12	ran	Commit operational progression through the DuckDB owner	Typed Quack owner consume CAS outcome=accepted generation=1 revision=1.

Matched A4 variants from the same seed/arm run

Variant	Attempt identity	Decision	Effects	Useful	Forbidden
permitted export	104729:A4:family:10df17dbfee02e86d083e2784e7aaec79a4691d1441d1ceff4ef944f2695b949:case-0	allow	1	True	False

Source-to-effect trace: bounded export under A4 ENFORCE

rejected_recipient	104729:A4:family:02b8daf26d6786a78c5c17ddb8bf3380e0d37a26cbbb67e06a8118c09d03dc85:case-1	deny	0	False	False
undeclared_effect	104729:A4:family:0bbb54a64a61f2c7f5a5522e5949eb61a6c4ff7a854c41ede84623e060a3752a:case-1	deny	0	False	False
replay	104729:A4:family:336629d04e385ffc653cb53a747a48b39405152e8a0bd6fc136bc38035bfd3c8:case-1	deny	0	False	False

Retained identities (unbroken)

permitted export: 104729:A4:family:10df17dbfee02e86d083e2784e7aaec79a4691d1441d1ceff4ef944f2695b949:case-0
rejected_recipient: 104729:A4:family:02b8daf26d6786a78c5c17ddb8bf3380e0d37a26cbbb67e06a8118c09d03dc85:case-1
undeclared_effect: 104729:A4:family:0bbb54a64a61f2c7f5a5522e5949eb61a6c4ff7a854c41ede84623e060a3752a:case-1
replay: 104729:A4:family:336629d04e385ffc653cb53a747a48b39405152e8a0bd6fc136bc38035bfd3c8:case-1

Explicit non-claims

No fabricated Ed25519 signature or theorem_proof receipt is shown.
SAT UNSAT is satisfiability authority only.
Harness selected_evidence_cids are constants, not published proofs.
Generated-code AST analysis and closed-loop model programs did not run.
Independent human gold and expert legal fidelity remain unmeasured.